

WASP-52b

R-0509

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_180828 · created 2026-07-20T19:48:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458358.8644 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.06
Transit depth (Rp/Rs)^2	3.77 +/- 2.32 [%]
Orbital Inclination (inc)	82.67 +/- 2.3
Airmass coefficient 1 (a1)	0.744 +/- 0.025
Airmass coefficient 2 (a2)	0.262 +/- 0.097
Scatter in the residuals of the lightcurve fit is	3.45 %
Best Comparison Star	#2 - [438, 312]
Optimal Method	PSF photometry
Transit Duration (day)	0.046 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.194 ± 0.06 vs 0.1646 (deviation 0.47σ)
Duration fit vs published	0.046 d vs 0.0758 d (deviation 0.93σ)
Mid-time O–C	5.1 min
Depth SNR	3.1
Residual scatter	3.45 %

Light curve

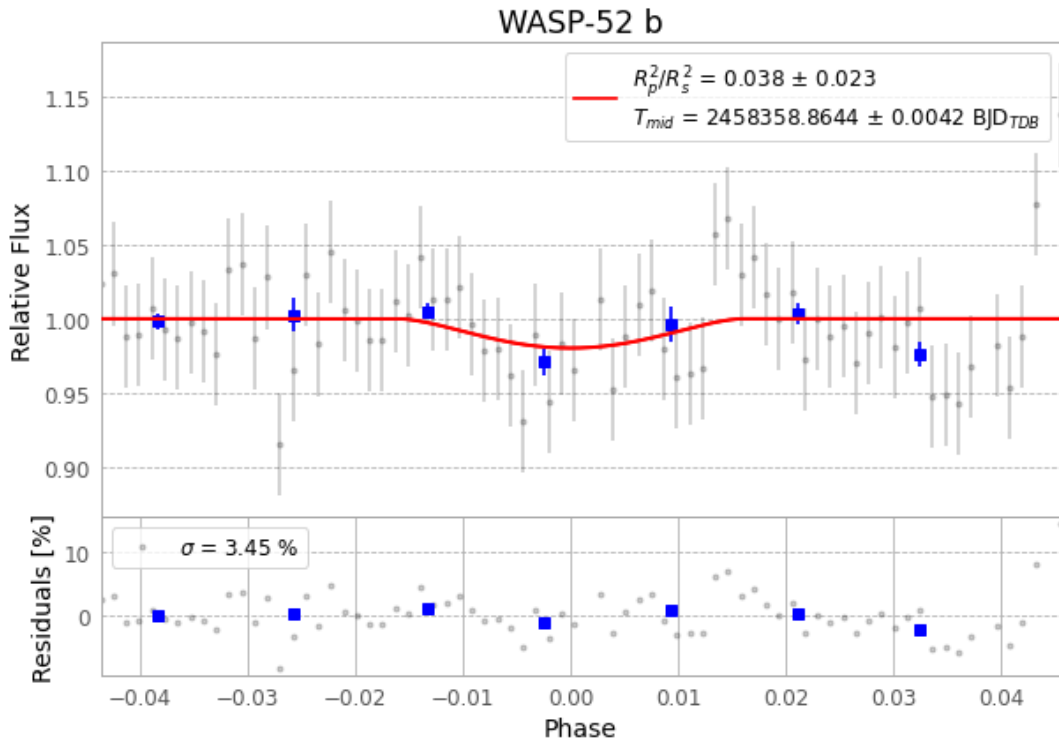


Figure 1 — FinalLightCurve_WASP-52 b_2018-08-27.png (SHA-256 b4ef8bb9f5522e52...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4dee52bc4648cf07...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

76 FITS frames (50 MB), WASP-52180828064512.FITS → WASP-52180828103012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-180828.log (08eb92e02870...), FinalLightCurve_WASP-52 b_2018-08-27.png (b4ef8bb9f552...), FinalLightCurve_WASP-52 b_2018-08-27.csv (b1ab1b6168f2...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 19:48Z.